12. DatabaseMetaData interface

The DatabaseMetaData interface in Java, part of the java.sql package, provides metadata about the database as a whole. It helps retrieve information about the database, its tables, supported features, and much more. This is particularly useful for developing database-agnostic applications.

Key Features of DatabaseMetaData

- Retrieve general information about the database (e.g., version, driver, name).
- Explore table and column details.
- Check supported SQL features and database-specific functionalities.
- Retrieve schema information for database introspection.

Key Methods in DatabaseMetaData

Method	Description
String getDatabaseProductName()	Retrieves the name of the database product.
String getDatabaseProductVersion()	Retrieves the version of the database product.
String getDriverName()	Retrieves the name of the JDBC driver used.
String getURL()	Retrieves the URL for this database.
String getUserName()	Retrieves the username used to connect to the database.
ResultSet getTables(...)	Retrieves a ResultSet of table metadata for a given schema or catalog.
ResultSet getColumns(...)	Retrieves metadata for columns in a specified table.
boolean supportsStoredProcedures()	Checks if the database supports stored procedures.
boolean supportsBatchUpdates()	Checks if the database supports batch updates.
boolean isReadOnly()	Checks if the database is read-only.

Example 1: Fetch General Database Information

```
package com.lokesh;
import java.sql.*;

public class DatabaseMetaDataExample {
    public static void main(String[] args) {
        String jdbcUrl = "jdbc:mysql://localhost:3306/organization";
        String username = "root";
        String password = "root";

        Connection connection = null;

        try {
            // Establish connection
            connection = DriverManager.getConnection(jdbcUrl, username, password);
            DatabaseMetaData metaData = connection.getMetaData();

            // Fetch database details
            System.out.println("Database Product Name: " +
```

```
                                                          metaData.getDatabaseProductName());
            System.out.println("Database Product Version: " +
                                                       metaData.getDatabaseProductVersion());
            System.out.println("JDBC Driver Name: " + metaData.getDriverName());
            System.out.println("Database URL: " + metaData.getURL());
            System.out.println("User Name: " + metaData.getUserName());
            System.out.println("Read-Only: " + metaData.isReadOnly());

        } catch (SQLException e) {
            e.printStackTrace();
        } finally {
            // Close connection
            try {
                if (connection != null) connection.close();
            } catch (SQLException e) {
                e.printStackTrace();
            }
        }
    }
}
```

Output:

```
Database Product Name: MySQL
Database Product Version: 8.0.31
JDBC Driver Name: MySQL Connector/J
Database URL: jdbc:mysql://localhost:3306/organization
User Name: root@localhost
Read-Only: false
```

Example 2: Retrieve Table Metadata

```
package com.lokesh;

import java.sql.*;

public class TableMetaDataExample {
    public static void main(String[] args) {
        String jdbcUrl = "jdbc:mysql://localhost:3306/organization";
        String username = "root";
        String password = "root";

        Connection connection = null;
        ResultSet tables = null;

        try {
            // Establish connection
            connection = DriverManager.getConnection(jdbcUrl, username, password);
            DatabaseMetaData metaData = connection.getMetaData();

            // Retrieve table metadata
            tables = metaData.getTables(null, null, "%", new String[]{"TABLE"});
            System.out.println("Tables in the database:");

            while (tables.next()) {
                System.out.println("Table Name: " + tables.getString("TABLE_NAME"));
                System.out.println("Table Type: " + tables.getString("TABLE_TYPE"));
                System.out.println("-----------------------------------------");
            }

        } catch (SQLException e) {
            e.printStackTrace();
```

```
        } finally {
            // Close resources
            try {
                if (tables != null) tables.close();
                if (connection != null) connection.close();
            } catch (SQLException e) {
                e.printStackTrace();
            }
        }
    }
}
```

Output:

```
Tables in the database:
Table Name: account
Table Type: TABLE
-----------------------------------------
Table Name: bankaccont
Table Type: TABLE
-----------------------------------------
Table Name: bhavana
Table Type: TABLE
-----------------------------------------
Table Name: blog_post
Table Type: TABLE
-----------------------------------------
Table Name: course
Table Type: TABLE
-----------------------------------------
Table Name: dateio
Table Type: TABLE
-----------------------------------------
Table Name: dateperform
Table Type: TABLE
-----------------------------------------
Table Name: employee
Table Type: TABLE
-----------------------------------------
Table Name: hibernate_sequence
Table Type: TABLE
-----------------------------------------
Table Name: jobseekar
Table Type: TABLE
-----------------------------------------
Table Name: lokesh
Table Type: TABLE
-----------------------------------------
Table Name: mobilephone
Table Type: TABLE
-----------------------------------------
Table Name: person
Table Type: TABLE
-----------------------------------------
Table Name: pic
Table Type: TABLE
-----------------------------------------
Table Name: register
Table Type: TABLE
-----------------------------------------
Table Name: skills
Table Type: TABLE
-----------------------------------------
Table Name: stocks
Table Type: TABLE
```

```
-----------------------------------------
Table Name: student
Table Type: TABLE
-----------------------------------------
Table Name: student_account
Table Type: TABLE
-----------------------------------------
Table Name: studentapp
Table Type: TABLE
-----------------------------------------
Table Name: upimage
Table Type: TABLE
-----------------------------------------
Table Name: user
Table Type: TABLE
-----------------------------------------
Table Name: user_table
Table Type: TABLE
-----------------------------------------
Table Name: users
Table Type: TABLE
-----------------------------------------
Table Name: datetimeexample
Table Type: TABLE
-----------------------------------------
Table Name: employee
Table Type: TABLE
-----------------------------------------
Table Name: actor
Table Type: TABLE
-----------------------------------------
Table Name: address
Table Type: TABLE
-----------------------------------------
Table Name: category
Table Type: TABLE
-----------------------------------------
Table Name: city
Table Type: TABLE
-----------------------------------------
Table Name: country
Table Type: TABLE
-----------------------------------------
Table Name: customer
Table Type: TABLE
-----------------------------------------
Table Name: film
Table Type: TABLE
-----------------------------------------
Table Name: film_actor
Table Type: TABLE
-----------------------------------------
Table Name: film_category
Table Type: TABLE
-----------------------------------------
Table Name: film_text
Table Type: TABLE
-----------------------------------------
Table Name: inventory
Table Type: TABLE
-----------------------------------------
Table Name: language
Table Type: TABLE
-----------------------------------------
Table Name: payment
Table Type: TABLE
```

```
-----------------------------------------
Table Name: rental
Table Type: TABLE
-----------------------------------------
Table Name: staff
Table Type: TABLE
-----------------------------------------
Table Name: store
Table Type: TABLE
-----------------------------------------
Table Name: bankaccont
Table Type: TABLE
-----------------------------------------
Table Name: jobseekar
Table Type: TABLE
-----------------------------------------
Table Name: mobilephone
Table Type: TABLE
-----------------------------------------
Table Name: tab
Table Type: TABLE
-----------------------------------------
Table Name: sys_config
Table Type: TABLE
-----------------------------------------
Table Name: city
Table Type: TABLE
-----------------------------------------
Table Name: country
Table Type: TABLE
-----------------------------------------
Table Name: countrylanguage
Table Type: TABLE
-----------------------------------------
```